

Carbosulfenylation of Alkenes with Organozinc Reagents and Dimethyl(methylthio)sulfonium Trifluoromethanesulfonate

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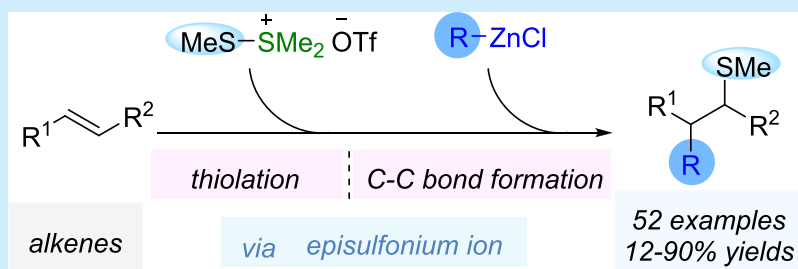
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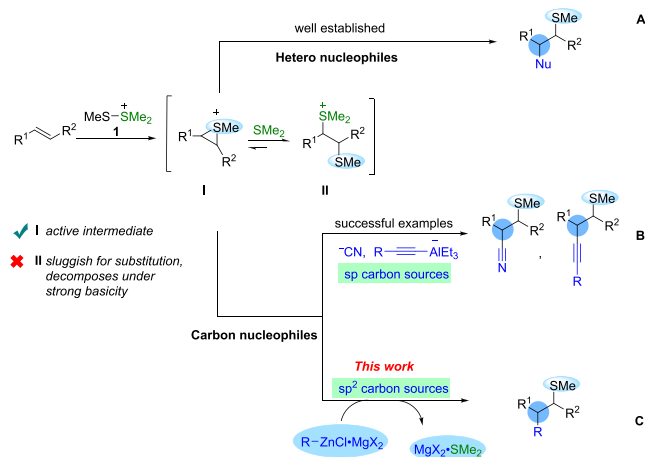
Supporting Information



ABSTRACT: The electrophilic alkylthiolation of alkenes, initiated by dimethyl(methylthio)sulfonium salts and the subsequent addition of various heteronucleophiles has been well-established. Regarding the use of carbon nucleophiles, however, only carefully designed sp -type carbon sources have been successfully applied. We herein present our findings on the methylthiolation of alkenes with dimethyl(methylthio)sulfonium trifluoromethanesulfonate, followed by carbon–carbon bond formation in the presence of organozinc reagents, thus achieving a catalyst-free protocol toward to the carbosulfenylation of alkenes.

Dimethyl(methylthio)sulfonium trifluoromethanesulfonate (DMTSM) and dimethyl(methylthio)sulfonium tetrafluoroborate (DMTSF)¹ are readily available, crystallizable, and easily handled electrophilic thiolating reagents. Their applications in the direct alkylthiolation of various nucleophiles,² initiation of dithioacetals to undergo the formation of thionium ion induced carbon–carbon bonds,³ and macrocyclization⁴ have been studied. Thioglycosides⁵ and peptides⁶ have also been found to undertake important conversions with DMTSM or DMTSF as the activator. Pioneered by the group of Trost, the electrophilic thiolation of alkenes and subsequent nucleophilic addition represents the most attractive application of these two dimethyl(methylthio)sulfonium salts in synthetic chemistry, because alkenes are commonly found in synthetic intermediates and industrial feedstocks. A vast scope of heteronucleophiles has been explored in this context, leading to the development of azasulfenylation,⁷ oxy(oxo)-sulfenylation,⁸ fluorosulfenylation,⁹ and other sulfenofunctionalizations¹⁰ of various alkenes (see Scheme 1A). Conceptually, in the presence of **I**, alkenes would deliver an equilibrium of active episulfonium ion **I** and sluggish thiosulfonium ion **II**. By submitting to the reported heteronucleophiles, intermediate thiosulfonium ion **II** can be persuaded to the conversion of its episulfonium analogue **I** prior to the formation of carbon–hetero bonds. Whereas, a typical nucleophilic carbon reagent with stronger basicity could result in fast decomposition of sulfonium salt **II**, thus excises anticipated carbon–carbon bond formation.¹¹ As a consequence, most of the typical carbon nucleophiles have proved problematic to undergo the

Scheme 1. State of the Art and Proposed Strategy



carbosulfenylation of alkenes initiated by DMTSF or DMTSM. Two successful examples—cyanosulfenylation and alkynylsulfenylation (Scheme 1B)—have been reported

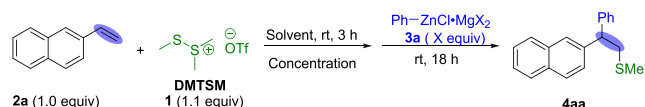
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by Trost group,^{8a,11a} both by employing *sp* carbon nucleophiles—cyanide and alkynylalaneate (prepared by the reaction of alkynyl lithium with triethylaluminum). The key to the success was interpreted as that these carbon nucleophiles were of low basicity, which allowed the conversion of intermediate **II** to **I**. Recent studies involving episulfonium ion intermediates¹² have introduced asymmetrical sulfenofunctionalization of alkenes,¹³ leading to the prosperity of imaginative protocols. However, a directly carbosulfenylation of alkenes with different types of carbon sources, initiated by dimethyl(methylthio)sulfonium salt, would still find academic significance and synthetic utility.

Organozinc reagents have been widely used as general nucleophilic partners in reactions of carbon–carbon bond formation and are readily available in laboratory synthesis.¹⁴ They have been found to possess lower basicity and excellent functional group tolerability, because of the relatively covalent carbon–zinc bond, in comparison with main group carbon–metal bonds such as carbon–lithium and carbon–magnesium.¹⁵ Although they generally were sluggish to undergo uncatalyzed reactions, they have been reported to react with a few reactive electrophiles, such as halogens, halonium ions, and phosphine chlorides.¹⁶ Furthermore, magnesium salts have been demonstrated to promote the reactions of organozinc reagents and electrophilic carbon partners in a catalyst-free fashion.¹⁷ We envision that magnesium salts would possibly be able to drive the equilibrium previously discussed to the episulfonium ion **I** by a strong coordination with dimethylthioether, thereby facilitating the successive carbon–carbon bond formation upon the addition of an organozinc reagent to the active intermediate **I** (Scheme 1C). Herein, we report the reaction that enables the access to sulfenyl carbon–carbon bond formation of various alkenes, by the addition of organozinc reagents in the presence of magnesium salt, with DMTSM as the activator.

We chose the sulfenyl arylation of 2-vinylnaphthalene **2a** as a model reaction and selected DMTSM (**1**) for the step of sulfonium ion generation. Quantitative conversion of alkene was observed by thin-layer chromatography (TLC), 3 h after mixing **2a** and slightly excess DMTSM in dichloromethane (DCM, 0.05 M) at ambient temperature. Successive addition of 1.5 equiv PhZnCl·MgCl₂ (**3a**) in THF (0.45 M, prepared from stoichiometric phenyl magnesium chloride and zinc chloride) to the reaction mixture afforded the sulfenyl arylated product **4aa** in very promising isolated yield (80%) after flash column chromatography (Table 1, entry 1). Satisfactory yield (89%) of **4aa** was obtained by increasing the chemical amount of organozinc and magnesium salt complex to 2.0 equiv. Further excess of **3a** (Table 1, entry 3) or slightly concentrated sulfonium ion intermediate (Table 1, entry 4) provided relatively lower yields. A diminished yield (48%) of target compound **4aa** was obtained when 1.0 equiv of LiCl was employed as an additive (Table 1, entry 5). Whereas, only a trace amount of **4aa** could be detected when phenyl zinc chloride prepared from phenyl lithium and zinc dichloride was applied (Table 1, entry 6), which was believed to be consistent with our suspect, that the magnesium salt could drive the equilibrium previously discussed to the formation of active intermediate **I** via the complexation with dimethylthioether, thus facilitating the formation of anticipated product **4aa**. Replacing DCM with dichloroethane (DCE) or polar solvents, including tetrahydrofuran (THF), acetonitrile (MeCN), CH₃NO₂, and N-methyl-2-pyrrolidone (NMP), leading to

Table 1. Optimization of the Conditions^a



entry	solvent	concentration (M)	X	yield ^b (%)
1	CH ₂ Cl ₂	0.05	1.5	71
2	CH ₂ Cl ₂	0.05	2.0	89
3	CH ₂ Cl ₂	0.05	2.5	50
4	CH ₂ Cl ₂	0.1	2.0	80
5 ^c	CH ₂ Cl ₂	0.05	2.0	48
6 ^d	CH ₂ Cl ₂	0.05	2.0	trace
7	ClCH ₂ CH ₂ Cl	0.05	2.0	65
8	THF	0.05	2.0	41
9	CH ₃ CN	0.05	2.0	26
10	CH ₃ NO ₂	0.05	2.0	trace
11	NMP	0.05	2.0	26
12 ^e	CH ₂ Cl ₂	0.05	2.0	trace
13 ^f	CH ₂ Cl ₂	0.05	2.0	trace
14 ^g	CH ₂ Cl ₂	0.05	2.0	trace

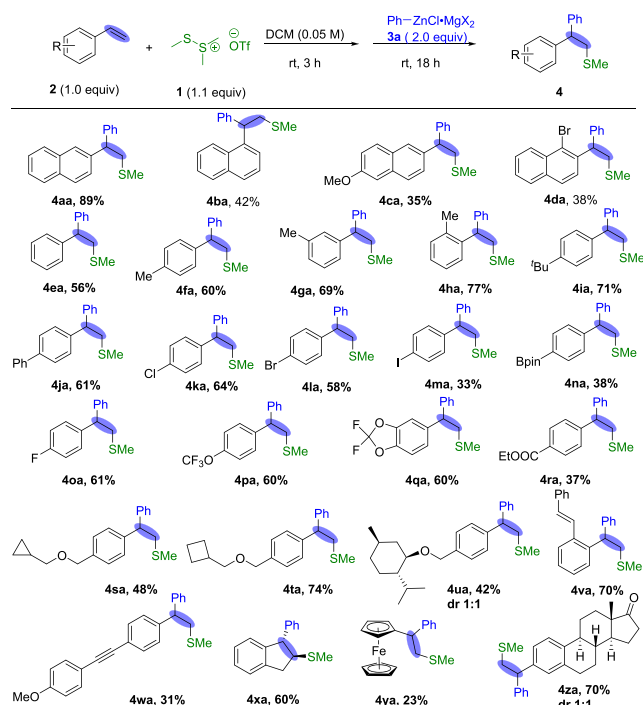
^aStandard conditions: alkene **2a** 0.27 mmol, DMTSM **1** 0.3 mmol, solvent 5 mL, PhZnCl **3a** 0.54 mmol (1.2 mL, 0.45 M in THF).

^bIsolated yields are given. ^c1.0 equiv of LiCl was added with **3a**.

^dPhZnCl was prepared from PhLi and ZnCl₂; ^e5 mol % Pd(PPh₃)₂Cl₂ was added immediately after the addition of phenyl zinc chloride solution. ^f5 mol % CuI was added immediately after the addition of phenyl zinc chloride solution. ^g5 mol % Ni(acac)₂ was added immediately after the addition of phenyl zinc chloride solution.

lower yields (Table 1, entries 7–11). Note that attempts to employ Pd(PPh₃)₂Cl₂, CuI, and Ni(acac)₂ as transition-metal catalysts for the stage of carbon–carbon bond formation led to the absence of anticipated product (Table 1, entries 12–14).

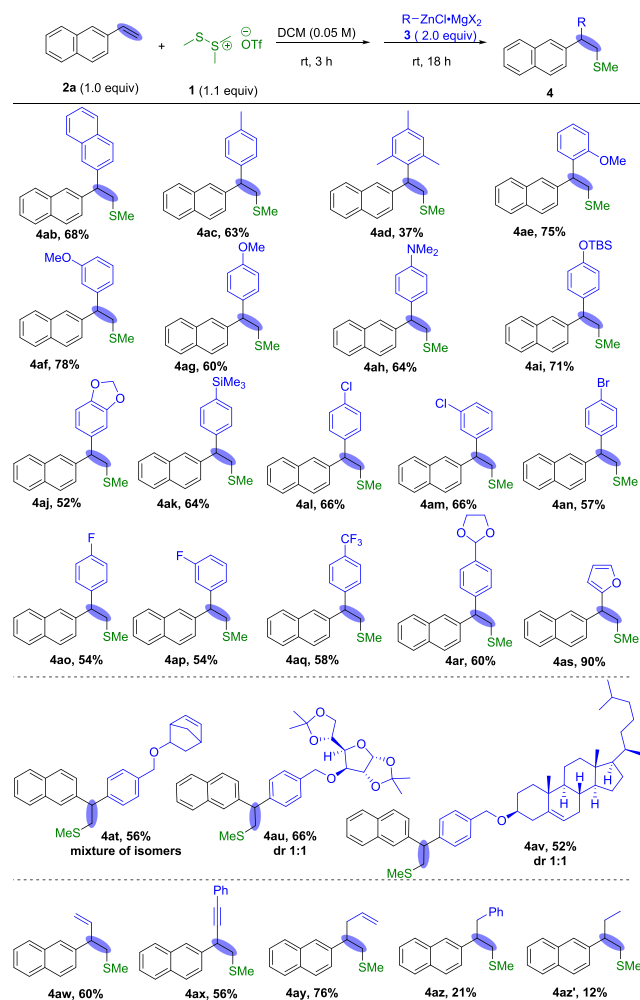
With optimized conditions in hand, we then examined the scope of alkenes for this protocol of aryl sulfenylation reaction. As shown in Scheme 2, a broad range of styrenes was able to deliver desired sulfide products. Both substituted naphthalene and phenyl rings were applicable to the reaction conditions (**4aa–4ea**). Styrenes with *ortho*-, *meta*-, *para*-alkyl or aryl groups were all good substrates for this protocol (**4fa–4ja**). Note that the tolerance of chloride, bromide, and iodide, as well as pinacol boronate ester, on the rings of styrenes indicated the compatibility with the well-developed transition-metal-catalyzed cross-coupling reactions of this methodology (**4ka–4na**), establishing its synthetic potential in the construction of thio-derived building blocks. Fluorine-containing styrenes were also amenable, affording the corresponding sulfenoarylated products in good yields (**4oa–4qa**). Interestingly, small-ring-containing styrenes were viable partners to this transformation with moderate to good yields of the corresponding products obtained (**4sa–4ua**). Substituted carbon–carbon multiple bonds were able to be delivered to the target products under the standard conditions (**4va** and **4wa**). Disubstituted styrene (indene) also smoothly underwent aryl sulfenylation under these conditions, delivering a moderate yield of product **4xa**. A *trans* conformation of **4xa** was evidenced by the X-ray diffraction analysis of its corresponding sulfone derivative (see the Supporting Information for details). Ferrocene-derived sulfide compound **4ya** was obtained when vinyl ferrocene was applied to this method. The possibility of applying estrone-derived styrene, furnishing the corresponding methylthio compound **4za** in 70% yield, further highlighted the generality of this protocol by

Scheme 2. Scope with Respect to Alkenes^a

^aStandard conditions: alkene **2**, 0.27 mmol; DMTSM **1**, 0.3 mmol; CH₂Cl₂, 5 mL; PhZnCl **3a**, 0.54 mmol (1.2 mL, 0.45 M in THF). Isolated yields are given. Bpin = pinacol boronate.

tolerating a relatively more polar C=O double bond. Arylated sulfide product was not found when α -methylstyrene was applied in this protocol, indicating that the carbon–carbon bond formation might be sensitive to steric hindrance. In addition, allylbenzene, cyclohexene, and cyclododecene were proved unsuccessful to undergo this conversion under the standard conditions. Note that this approach of carbosulfenylation showed excellent selectivity, with the formal Markovnikov addition being predominant.

Scheme 3 depicts the scope of the sulfonyl arylation reaction, with respect to the scope of arylzinc reagents. Both naphthalene and toluene ring-derived zinc compounds were amenable, providing sulfide products in good yields (**4ab** and **4ac**). Very bulky units also were able to be delivered into the anticipated product (**4ad**). Electron-rich aryl zinc reagents, methoxyl- and *N,N*-dimethylamino-phenyl zinc chlorides, all furnished the target thioethers in reasonable yields (**4ae**–**4ah**). Protected phenol and diphenol derived products were ready to be synthesized under the standard conditions (**4ai** and **4aj**). Potential Hiyama cross-coupling thio-containing partner **4ak** could be prepared by using trimethylsilane (TMS)-substituted arylzinc chloride. Chloride and bromide functionalities were also able to be delivered from the aryl zinc nucleophiles (**4al**–**4an**). Fluorine-containing arylzinc reagents were suitable reaction partners for this protocol (**4ao**–**4aq**). Ethylene-glycol-protected aldehyde-derived aryl zinc was also applicable and provided the corresponding sulfide product in synthetically useful yield (**4ar**). Excellent yield (90%) of carbosulfenylated product (**4as**) was obtained when furan-derived organozinc was subjected to the standard conditions. As it is well-known that functionalized organozinc reagents can be readily prepared by Mg/I exchange and the following metathesis reaction with zinc dichloride,¹⁸ we then prepared arylzinc

Scheme 3. Scope with Respect to Arylzinc Reagents^a

^aStandard conditions: alkene **2a**, 0.27 mmol; DMTSM **1**, 0.3 mmol; CH₂Cl₂, 5 mL; and RZnCl **3**, 0.54 mmol (1.2 mL, 0.45 M in THF). Isolated yields are given.

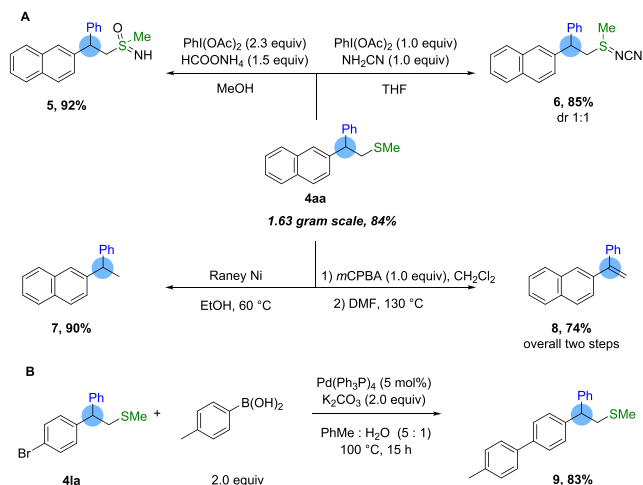
chlorides derived from norbornene, α -D-glucufuranose and cholesterol derivatives using the Knochel procedure, and we were delighted to find that they all reacted with the sulfonium ion intermediate smoothly, furnishing the anticipated structures in satisfactory yields (**4at**–**4av**). However, attempts to apply electron-deficient hetero arylzincs, pyrazin-2-ylzinc chloride, and pyridin-2-ylzinc chloride to this method led to no anticipated products detected.

Encouraged by the excellent reactivity of the sulfonium ion intermediate toward arylzinc reagents, we then tried to apply several other types of organozinc reagents in this sequence (see **Scheme 3**, **4aw**–**4az'**). Another typical sp² carbon nucleophile, vinyl zinc chloride, behaved similar to its phenyl analogues, affording arylvinyl thioether product **4aw** in 60% yield. Relatively stable acetylenyl zinc chloride was also proven compatible with our standard conditions (**4ax**). The allylzinc reagent underwent this transformation smoothly, delivering the target product **4ay** in good yield. Relatively lower yields of anticipated products were detected when alkyl zinc nucleophiles, benzyl zinc chloride and ethyl zinc chloride, were subjected to the standard conditions (**4az** and **4az'**). Competitive product **S**_{4az} (see the **Supporting Information**),

which arose from the dimethylthiolation of 2-vinylnaphthalene, could be isolated from these two reactions.

Moreover, performing the reaction in gram scale facilitated the synthesis of target compound **4aa** in 84% yield (Scheme 4A). The synthetic utility of our sulfide product was then

Scheme 4. Synthetic Applications^a



^a(A) Compound **4aa** can be prepared in gram scale (transformations of compound **4aa** are shown); (B) palladium-catalyzed cross-coupling of **4la** with 4-tolylboronic acid.

outlined by the transformations of product **4aa**. Treated with (diacetoxyiodo)benzene and ammonium carbonate in methanol, **4aa** was converted to the corresponding sulfoximine **5** in excellent yield.¹⁹ N-cyano sulfilimine **6** could be readily prepared by an imination procedure with cyanogen amine in the presence of PhI(OAc)₂.²⁰ Subjecting **4aa** to the mixture of Raney-Ni in ethanol led to the removal of sulfide residue, providing 1,1-diarylethane product **7** in excellent yield. The methylthio derivative could be readily oxidized by stoichiometric *meta*-chloroperoxybenzoic acid (*m*-CPBA), delivering the corresponding methylsulfoxyl analogues, in which the methylsulfoxyl group is easily removed by elevating the temperature of its solution in DMF, with 1,1-diarylethylene **8** as the product. Furthermore, a palladium-catalyzed cross-coupling of our sulfide product **4la** with tolylboronic acid was conducted,²¹ leading to the synthesis of biaryl sulfide compound **9** in 83% yield (Scheme 4, B). These nice derivations of sulfide product illustrated the versatility and possible power in laboratory syntheses of the protocol presented herein.

In summary, a reaction of catalyst-free sulfenyl carbon–carbon bond formation on alkenes has been developed, by employing DMTSM as the initiator at the activation stage. Benefiting from the strategic application of readily available organozinc reagents as the nucleophilic carbon sources, this process shows excellent practicality and generality by tolerating the large scope of functional groups within both alkene and organozinc partners. We believe that this simple and modular methodology should find many adoptions in synthetic chemistry.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.0c03810>.

Experimental details, characterization data, and spectra (PDF)

Accession Codes

CCDC 2033474 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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